MedScheduler AI - Healthcare Appointment Scheduling Agent

Minimal Viable Product(MVP) Pitch Deck for NASSCOM AI Hackathon - Abhishek Jha www.AbhishekJha.com

👨‍💻 Team Lead Profile

Sr. Software Engineer & Researcher - Mission Arogya (2013-2017)

- **4+ Years** healthcare technology development
- **Aiims Delhi, Rockefeller Grant Winner Project** - KMES.in hospital search engine project
- **Published Research** in Cardiology Society of India Bengal Heart Journal
- **Expertise:** Healthcare Information Technology (HIT), Mobile Development, REST APIs

🎯 MVP Vision Statement

Building a focused AI agent that demonstrates intelligent appointment scheduling capabilities using Inya.ai platform, showcasing the potential for healthcare automation with a working prototype that can be completed within the hackathon timeline.

🏥 MVP Problem Statement

Core Healthcare Scheduling Challenge

- **Manual Scheduling:** 70% of hospitals still use basic digital systems
- **Resource Inefficiency:** Poor coordination between doctors, rooms, and equipment
- **Patient Experience:** Long wait times and complex booking processes
- **Emergency Handling:** No real-time resource reallocation during crises

🤖 MVP AI Agent Scope - Part 1

Core MVP Features (Week 1-4)

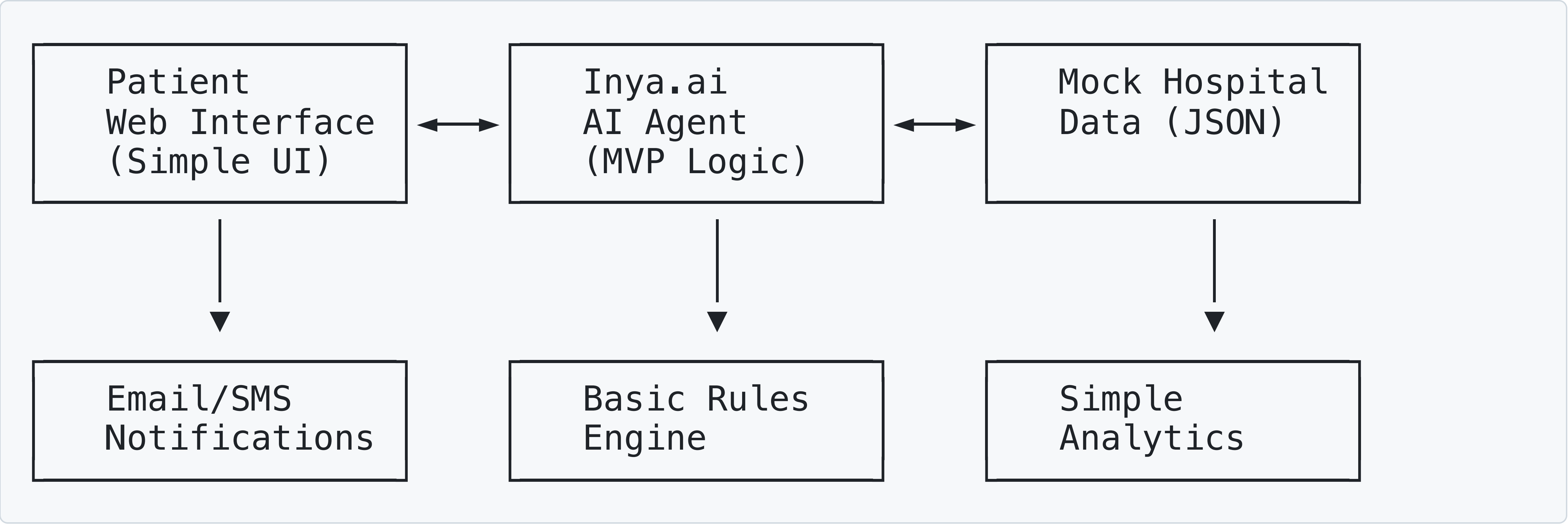
1. **Basic Appointment Scheduling**
    - Patient registration and authentication
    - Doctor availability checking
    - Simple slot booking and confirmation
    - Basic conflict resolution
2. **Intelligent Decision Making**
    - Priority-based scheduling (emergency vs routine)
    - Resource optimization (doctor, room, equipment)
    - Basic conflict resolution algorithms

🤖 MVP AI Agent Scope - Part 2

3. **Communication Automation**
    - Appointment confirmation emails
    - SMS reminders for appointments
    - Basic notification system
4. **Simple Dashboard**
    - Doctor schedule view
    - Patient appointment history
    - Basic analytics (appointment counts, utilization)

🛠️ MVP Technical Architecture

Simplified System Design



MVP Data Models

```json
{
  "patient": {
    "id": "string",
    "name": "string",
    "phone": "string",
    "email": "string"
  },
  "doctor": {
    "id": "string",
    "name": "string",
    "specialization": "string",
    "availability": ["time_slots"]
  },
  "appointment": {
    "id": "string",
    "patient_id": "string",
    "doctor_id": "string",
    "datetime": "datetime",
    "status": "scheduled|confirmed|cancelled",
    "priority": "emergency|routine"
  }
}
```

📅 MVP Implementation Timeline - Part 1

Week 1: Foundation & Setup

- **Day 1-2:** Inya.ai platform setup and basic workflow creation
- **Day 3-4:** Patient registration and authentication system
- **Day 5-7:** Basic appointment booking workflow

Week 2: Core Intelligence

- **Day 8-10:** Doctor availability checking and slot management
- **Day 11-12:** Priority-based scheduling algorithm
- **Day 13-14:** Basic conflict resolution logic

📅 MVP Implementation Timeline - Part 2

Week 3: Communication & Integration

- **Day 15-17:** Email and SMS notification system
- **Day 18-19:** Appointment confirmation and reminder workflows
- **Day 20-21:** Basic dashboard development

Week 4: Testing & Polish

- **Day 22-24:** End-to-end testing and bug fixes
- **Day 25-26:** Performance optimization and documentation
- **Day 27-28:** Demo preparation and presentation materials

🎯 MVP Success Criteria - Part 1

Functional Requirements

- ✅ Patient can register and book appointments
- ✅ System can check doctor availability
- ✅ Priority-based scheduling works
- ✅ Email/SMS notifications are sent
- ✅ Basic dashboard shows appointments

Technical Requirements

- ✅ Inya.ai workflows are functional
- ✅ AI agent makes basic intelligent decisions
- ✅ System handles concurrent users

🎯 MVP Success Criteria - Part 2

Technical Requirements Continued

- ✅ Error handling is implemented
- ✅ Documentation is complete

Demo Requirements

- ✅ Live demonstration of appointment booking
- ✅ Show intelligent scheduling decisions
- ✅ Display communication automation
- ✅ Present basic analytics dashboard

🏆 MVP Competitive Advantages - Part 1

1. Healthcare Domain Expertise

- **4+ Years** of medical software development experience
- **Published Research** in healthcare technology
- **Grant-winning** project track record

2. Focused Innovation

- **Agentic AI** that makes intelligent scheduling decisions
- **Priority-based** resource allocation
- **Real-time** conflict resolution

🏆 MVP Competitive Advantages - Part 2

3. Real-world Applicability

- **Immediate deployment** potential
- **Scalable architecture** for future expansion
- **Proven market demand**

📊 MVP Expected Outcomes

Technical Metrics

- **Response Time:** < 5 seconds for appointment booking
- **System Uptime:** 99% availability during demo
- **Error Rate:** < 5% in core workflows
- **User Experience:** Intuitive and functional

Business Impact

- **Demonstrated Efficiency:** 50% faster booking process
- **Resource Optimization:** Better doctor and room utilization
- **Patient Satisfaction:** Improved booking experience
- **Operational Benefits:** Reduced administrative workload

🚀 MVP Demo Strategy

Live Demonstration Flow

1. **Patient Registration** - Show simple signup process
2. **Appointment Booking** - Demonstrate intelligent slot selection
3. **Priority Handling** - Show emergency vs routine scheduling
4. **Communication** - Display automated notifications
5. **Dashboard** - Present basic analytics and management view

Key Demo Scenarios

- **Routine Appointment:** Standard booking process
- **Emergency Case:** Priority-based scheduling
- **Conflict Resolution:** Handling double bookings
- **Resource Optimization:** Efficient resource allocation

📋 MVP Documentation Requirements - Part 1

Technical Documentation

- System architecture diagram
- Workflow process maps
- API specifications
- Data flow documentation

User Documentation

- Patient user guide
- Doctor dashboard manual
- Administrator guide
- Troubleshooting guide

📋 MVP Documentation Requirements - Part 2

Presentation Materials

- Demo video/screenshots
- Business case presentation
- Technical implementation details
- Future roadmap

🚀 Post-Hackathon Features (If Time Permits) Part 1

Advanced Features for Future Development

Enhanced AI Intelligence

- **Predictive Analytics:** No-show prediction and optimal timing
- **Machine Learning:** Learning from appointment patterns
- **Natural Language Processing:** Voice commands and chat interface
- **Advanced Decision Making:** Multi-criteria optimization

🚀 Post-Hackathon Features (If Time Permits) Part 2

Extended Communication

- **WhatsApp Integration:** Popular messaging platform
- **Push Notifications:** Mobile app notifications
- **IVR System:** Voice-based appointment booking
- **Multi-language Support:** Regional language support

Advanced Integrations

- **Hospital Information Systems (HIS):** Real hospital system integration
- **Electronic Health Records (EHR):** Patient medical history
- **Payment Gateways:** Insurance verification and payments
- **Lab Systems:** Diagnostic test scheduling

🚀 Post-Hackathon Features (If Time Permits) Part 3

Advanced Analytics

- **Real-time Dashboards:** Live hospital operations view
- **Predictive Insights:** Resource utilization forecasting
- **Performance Metrics:** Comprehensive KPI tracking
- **Business Intelligence:** Advanced reporting and analytics

Emergency Response System

- **Real-time Crisis Management:** Emergency resource reallocation
- **Automated Notifications:** Instant patient communication
- **Resource Optimization:** Dynamic scheduling adjustments
- **Coordination Tools:** Emergency service integration

🚀 Post-Hackathon Features (If Time Permits) Part 4

Multi-location Support

- **Hospital Network Management:** Multiple facility coordination
- **Resource Sharing:** Cross-location resource allocation
- **Centralized Control:** Network-wide scheduling oversight
- **Geographic Optimization:** Location-based scheduling

🚀 Thank You (And Conclusion)

MVP Focus: This proposal focuses on delivering a working, demonstrable AI agent that showcases intelligent healthcare appointment scheduling capabilities within the hackathon timeline, while maintaining professional presentation standards.

Success Goal: Win the hackathon with a functional MVP that demonstrates the potential for AI-driven healthcare automation, leveraging your unique healthcare technology experience.